

1 (a) time

B2

(electric) current

allow amount of substance

allow luminous intensity

any two of the above quantities, 1 mark each

(b) (i) $T = 2\pi \times \sqrt{\frac{200 \times 10^{-3}}{25}}$

$$= 0.562 \text{ s}$$

A1

$$\text{percentage uncertainty} = (2\% + 8\%) / 2 (= 5\%)$$

C1

or

$$\text{fractional uncertainty} = (0.02 + 0.08) / 2 (= 0.05)$$

$$\Delta T = 0.562 \times 0.05$$

$$= 0.028 \text{ s}$$

C1

$$T = (0.56 \pm 0.03) \text{ s}$$

A1

(ii) 1. total energy = max kinetic energy

$$= \frac{1}{2} (200 \times 10^{-3}) (2\pi / T)^2 (5.00 \times 10^{-2})^2$$

M1

$$= \frac{9.87 \times 10^{-3}}{T^2}$$

A1

$$\text{2. total energy} = \frac{9.87 \times 10^{-3}}{0.56^2}$$

$$= 3.15 \times 10^{-2} \text{ J}$$

A1

2

2 (a) sum / total momentum of bodies is constant

R1